Hamid Rizvi

AI Engineer | New York, NY | (929) 420-6592 | hamidrizvi@stern.nyu.edu | linkedin.com/in/hamid-rizvi | github.com/hamidrizvi4 | hamidrizvi.vercel.app

Summary

AI engineer who builds production LLM systems around a single principle: model output is untrusted input. I design the boundary between what an LLM decides and what deterministic, auditable code decides, then verify it with evals, threat models, and real measurements rather than assumptions.

Experience

LexTrack AI (1-Click Full Stack App Builder) — New York, NY
AI/ML Product Engineering Intern — *Feb 2026 – May 2026*

- Designed LLM-as-judge evals and a deterministic testing framework to benchmark GPT-4, Claude, and Gemini, setting the performance and cost criteria that guided production model selection
- Built and tuned production RAG systems and a 3-layer fallback architecture in TypeScript, cutting LLM inference costs 40% while holding sub-2s p95 latency across many B2B deployments at 99.5% uptime

Projects

ResidenceOS | *NYC Property Compliance Platform* — Aug 2026

- Found that NYC Local Law 152 deadlines follow DOB's rotating community-district schedule, not date + 4 years. The naive implementation would have produced the wrong legal deadline for two of three seeded buildings; built the arithmetic deterministically in TypeScript as versioned config, failing closed on missing district data
- Wrote a threat model treating LLM output as untrusted input: schema-validated inert extraction proposals gated behind human review, prompt-injection signals surfaced rather than dropped, plus Stripe webhook signature verification, replay idempotency, and secret redaction

Claude API Support Triage Agent | *Developer Support Triage and Routing Agent* — Jul 2026

- Built a 5-tool RAG agent on a raw tool-calling loop (no framework), where a rules-based routing engine overrides model confidence to force human review on billing, security, sales, and bug tickets, with every tool call logged as an audit trail
- Ran a 7-point audit that surfaced 12 findings including a live RLS misconfiguration exposing all 6 tables to unauthenticated read/write; measured free vs. paid models on an identical 40-ticket set, cutting incomplete runs from 50% to 7.5% for $0.25 verified against billing data

QuadTax | *AI Tax Compliance Engine* — Mar 2026

- Architected a strict split between LLM tasks (document parsing, classification) and deterministic financial logic, keeping hallucinations out of tax outputs and achieving 100% mathematical accuracy
- Built a Next.js/FastAPI app with a Tesseract/pdfplumber OCR pipeline and OpenAI Structured Outputs across W-2, 1042-S, and I-94 forms, cutting filing time from 35 to under 10 minutes

LexTrack AI | *Capstone Project, Lead AI Developer & Product Manager* — Dec 2025

- Built a hybrid architecture pairing deterministic rules with a Gemini LLM for a 0→1 reservation product, reaching 1–2s response times and 70% cost reduction via prompt engineering, caching, and fallback logic
- Shipped with 3 engineers and 1 designer to 50+ active users in 8 weeks, cutting merchant onboarding 80% (30min→5min) at 85% weekly retention

Education

New York University — New York, NY
M.S. in Management of Technology | GPA: 3.75/4.0 — *Sep 2024 – May 2026*

S.R.M. Institute of Science and Technology — Chennai, India
B.Tech in Computer Science Engineering w/ Cloud Computing — *Sep 2020 – May 2024*

Technical Skills

AI/LLM Systems: LLM Integration (GPT-4, Claude, Gemini), Agent Tool-Calling Loops, RAG, LLM-as-Judge Evals, Model Benchmarking, Prompt Engineering, Vector Databases, Hybrid AI Architecture, Cost Optimization, Prompt-Injection Mitigation

Engineering: Python, TypeScript, JavaScript, Next.js, React, Node.js, FastAPI, RESTful APIs, Supabase/Postgres, Row-Level Security, Stripe Webhooks, Fallback Architecture, Git/GitHub, CI/CD, Docker, AWS

Data & Testing: SQL, Pandas, NumPy, Pytest, OCR (Tesseract, pdfplumber), ETL Pipelines, Threat Modeling, Regression Testing, A/B Testing